LITERATURE SURVEY

Application of Quantum Computing in Digital Education Systems

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1. Introduction

Our project proposes a hybrid system for digital education. The classical computer handles student data and checks the rules, and the quantum part (QAOA) helps find a good personalized learning path by solving a QUBO model. For this survey we studied the four research papers selected by our team. Two of them look at quantum computing for education in general, one proposes a simulated e-learning system, and one studies research trends and tools.

2. Papers Reviewed

2.1 Application of Quantum Computing in Digital Education Systems

N. S. Gaddapuri, Power System Protection and Control, vol. 50, no. 2, pp. 12-24, 2022.

Problem statement: Digital education platforms create huge amounts of student data, and tasks like personalized learning paths, performance prediction and resource recommendation are hard to compute at scale. The paper notes that the number of possible learning sequences grows extremely fast.

Quantum approach: A hybrid framework. The classical side stores student profiles, the content library and the prerequisite graph. QAOA solves learning-path selection as a QUBO problem, and the classical side then checks prerequisites and ranks the paths. The paper also discusses quantum annealing for curriculum design, quantum SVM for prediction and quantum sampling for recommendation. Everything was tested by simulation on classical computers.

Findings: In simulations, quantum methods found 15-25% better learning paths than a classical greedy method for problems with 50-100 learning objectives, but only for complex problems. Smaller problems showed little benefit. The authors also admit that simulations cannot prove the same result on real quantum hardware.

Relevance: This is the closest paper to our project, and our hybrid architecture follows the same idea. It is also where the 15-25% figure in our abstract comes from. Since the paper does not share its datasets or code, we treat this number as an expected result that we still need to test ourselves.

2.2 Quantum Algorithms for Enhanced Educational Technologies

B. Hanafi, M. Ali and D. Singh, Discover Education, vol. 4, article 8, 2025.

Problem statement: Educational technology based on classical computing struggles with personalization, data security and resource management. The paper asks how quantum algorithms could help.

Quantum approach: It is a literature-based study. It explains algorithms such as Grover's search, quantum Fourier transform, quantum annealing, quantum SVM and quantum decision trees, with simple pseudocode, and links each to an education use. Quantum key distribution is suggested for protecting student data. For learning paths and scheduling, it points to quantum annealing.

Findings: The paper concludes that quantum methods could improve personalization, scheduling and data security. It also lists the obstacles: unstable hardware, noise, few qubits, the need for special skills, difficulty fitting into existing platforms, and privacy and fairness concerns. It recommends hybrid systems and cloud access to quantum hardware.

Relevance: It supports our choice of a hybrid design and of optimization methods for learning paths, and its list of challenges helps with our limitations. However, it is mostly theoretical and has no experiments of its own, so we cannot use it as proof of improvement.

2.3 Quantum Computing-Driven Mobile E-Learning Systems for Ultra-Secure and High-Speed Content Delivery

M. Ernazarova et al., Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications (JoWUA), vol. 17, no. 2, pp. 36-48, 2026.

Problem statement: Mobile e-learning depends on wireless networks and the cloud, which brings problems with data security, delay, scalability and reliable content delivery.

Quantum approach: The paper proposes QCMELS, a five-layer system covering login, quantum security, resource optimization, wireless delivery and cloud-edge storage. Since real quantum hardware is not available in such networks, the quantum key distribution is simulated with Qiskit and the optimization is quantum-inspired. The bulk content is encrypted with a normal classical method.

Findings: In simulation, QCMELS reached 97.8 Mbps throughput, 61 ms latency, 98.6% packet delivery and 98.1% security accuracy, which was better than four other mobile learning methods. The ablation study showed that removing the quantum security layer or the edge layer made results worse.

Relevance: It shows a layered hybrid design and the use of Qiskit simulation when hardware is not available, which is what we will also do. But it is about secure delivery and not learning paths, and its results come from simulated, quantum-inspired parts, so they do not prove a real quantum advantage.

2.4 Quantum Computing in Data Science and STEM Education: Mapping Academic Trends and Analyzing Practical Tools

E. Lopez-Meneses, J. Caceres-Tello, J. J. Galan-Hernandez and L. Lopez-Catalan, Computers, vol. 14, no. 6, article 235, 2025.

Problem statement: Quantum computing is growing fast, but few studies connect it with data science and teaching. The paper wants to know how the research is developing and which tools are suitable for teaching it.

Quantum approach: This is not a quantum algorithm paper. It is a bibliometric study of 281 Scopus papers from 2015 to 2024 using VOSviewer, along with a comparison of educational platforms such as Qiskit, Amazon Braket, Azure Quantum and Qinterpreter.

Findings: Research on quantum computing and education has grown quickly since 2020, but nearly half of it is conference papers and most comes from computer science and engineering. Research groups are scattered and teaching-focused work is limited. Qiskit stood out for its open access, documentation and use in universities. The authors also suggest a six-phase roadmap for bringing quantum content into university programs.

Relevance: It shows that this field is still young, which is a reason to be careful about big claims. It also supports using Qiskit for our QAOA simulation. Its relevance to our project is indirect.

3. Comparison

Paper	Focus	Type of evidence	Main limitation
Gaddapuri (2022)	Learning paths, curriculum, prediction, recommendation	Classical simulation	No named datasets or code; no real quantum hardware
Hanafi et al. (2025)	Quantum algorithms for personalization, security, resources	Literature review and theory	No experiments of its own
Ernazarova et al. (2026)	Secure and fast mobile e-learning delivery	Simulation with Qiskit	Quantum parts are simulated; not about learning paths
López-Meneses et al. (2025)	Research trends and tools for quantum education	Bibliometric analysis	About teaching quantum, not applying it

4. Gap and Conclusion

Among these four papers, only Gaddapuri's paper looks at learning-path optimization with QAOA and QUBO, and it does so as a framework tested by simulation. The other papers stay theoretical, focus on secure delivery, or study how quantum computing is taught. None of them runs QAOA on a learning-path dataset in a way that can be checked against classical baselines. That is the gap our project works on.

From this survey we understood that a hybrid approach is the practical choice for now, that most reported benefits come from simulation or theory, and that our expected 15-25% improvement must be tested against classical methods such as greedy, genetic and integer programming.

References

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